



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences

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ASSESSMENT TOOL -2

CO4_AT1

Name : S Nikhileswar Reddy

Register No : 192525096

Course code : CSA1613

Course name: Data Warehousing and Data Mining

Faculty name: Dr.Kanimozhi KV

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Data Warehousing and Data Mining COURSE CODE: CSA1613

CO4: Clustering Algorithms

Q1. Hospital Patient Segmentation — K-Means Elbow Method

Given Data

Patient ID	Age	Monthly Medical Expense (₹)	Health Risk Score
P1	28	12000	65
P2	55	25000	40
P3	42	20000	75
P4	30	11000	70
P5	60	28000	35

K	SSE
1	600
2	350
3	220
4	200
5	190

Interpretation of the Elbow Point

Plotting K against SSE shows the drop in error at each step: from K=1 to K=2 SSE falls by 250 (600→350), from K=2 to K=3 it falls by 130 (350→220), from K=3 to K=4 it falls by only 20 (220→200), and from K=4 to K=5 it falls by just 10 (200→190). The rate of decrease is steep up to K=3 and then flattens sharply, forming a visible "elbow" or bend in the curve at K=3. Beyond this point, adding more clusters yields diminishing returns — SSE keeps falling only marginally while model complexity increases.

Optimal Number of Clusters

The optimal number of clusters is K = 3. This is the point where the SSE curve bends most sharply, indicating that three clusters explain most of the variance in the data without over-fitting into unnecessarily granular groups.

Effect on Patient Segmentation and Healthcare Decision-Making

- Cluster 1 (Low expense, low-to-moderate risk): younger patients such as P1 and P4, who need routine preventive care and wellness monitoring.
- Cluster 2 (High expense, high risk): patients such as P3, needing intensive/critical care, priority resource allocation and closer monitoring.
- Cluster 3 (High expense, low risk): patients such as P2 and P5, whose high spending is not matched by proportionate risk — useful for cost-optimization and fraud/anomaly review.

Choosing $K=3$ lets the hospital design three distinct care pathways instead of treating all patients uniformly. Too few clusters ($K=1$ or 2) would merge clinically different patient groups and dilute targeted interventions, while too many clusters ($K=4$ or 5) would fragment the population into groups too small and too similar to justify separate treatment protocols, increasing administrative overhead without improving care quality. Hence $K=3$ gives the best balance between model simplicity and clinically meaningful segmentation, supporting better resource planning, personalized treatment plans, and cost management.

Q2. E-Commerce Order Analysis — Hierarchical Clustering / Dendrogram

Given Data

Order ID	Product Category	Order Value (₹)
O1	Electronics	55000
O2	Clothing	25000
O3	Electronics	57000
O4	Furniture	30000
O5	Clothing	27000

Interpretation of Cluster Formation

In agglomerative hierarchical clustering, each order starts as its own cluster and the two closest orders (by category and order value) are merged repeatedly until one cluster remains, recording each merge as a node in the dendrogram. O1 (₹55,000) and O3 (₹57,000) are both Electronics with very close order values, so they merge first at a very low distance. O2 (₹25,000) and O5 (₹27,000) are both Clothing with close values, so they merge next at a similarly low distance. O4 (Furniture, ₹30,000) sits closer in value to the Clothing pair than to the Electronics pair, so it merges with the Clothing–O5–O2 group at a

moderate distance. Finally, the Electronics cluster (O1, O3) merges with the Clothing+Furniture cluster at the highest distance, forming the root of the dendrogram.

Appropriate Number of Clusters

Cutting the dendrogram at a moderate height (below the final root merge but above the early low-distance merges) yields three clusters: {O1, O3} — high-value Electronics orders; {O2, O5} — low-value Clothing orders; and {O4} — Furniture, which is closer to the Clothing group in value but distinct in category. If the analyst prioritizes value over category, cutting slightly higher merges O4 with {O2, O5}, giving two clusters (High-value Electronics vs. Low/Mid-value Clothing & Furniture). Three clusters is generally the most informative cut here, since it keeps category identity intact while respecting the value proximity within categories.

Significance for Understanding Customer Purchasing Behavior

- The Electronics cluster reveals a high-average-order-value segment, useful for premium loyalty programs and upselling.
- The Clothing cluster reveals a low-value, high-frequency segment, useful for volume-based promotions and bundling offers.
- The Furniture order's position between clusters signals a transitional or occasional big-ticket buyer, useful for targeted cross-category marketing.

This segmentation helps the retailer design category-specific marketing, inventory, and pricing strategies, and the dendrogram's hierarchy also lets management view customer behavior at multiple levels of granularity (two broad segments or three finer segments) depending on the business need.

Q3. Mobile App User Behavior — Soft Clustering / Fuzzy Membership

Given Data

User	Daily Usage Time (min)	Login Frequency	In-App Purchase (₹)
U1	120	18	2000
U2	60	5	5000
U3	140	22	1500
U4	50	4	6000
U5	130	20	1800

Interpretation of Probability-Based Cluster Assignment

Using a soft-clustering approach such as Fuzzy C-Means (rather than hard HAC assignment), each user is not forced into exactly one cluster; instead every user receives a membership value (between 0 and 1, summing to 1 across clusters) indicating the degree to which it belongs to each cluster. Based on usage time and login frequency, two natural centers emerge: a High-Engagement group (U1, U3, U5 — high usage time 120-140 minutes, high login frequency 18-22, lower per-visit purchase amounts) and a Low-Engagement/High-Spender group (U2, U4 — low usage time 50-60 minutes, low login frequency 4-5, but high in-app purchase values 5000-6000). A user like U1 would have a high membership (e.g., ~0.85) toward the High-Engagement cluster and a low membership (~0.15) toward the other cluster, whereas a borderline user with moderate usage and moderate spend would show a more balanced split (e.g., 0.55/0.45), reflecting genuine ambiguity rather than an arbitrary hard cut.

Advantages of Soft Clustering over Hard Clustering

- Captures ambiguity: users whose behavior straddles two segments are represented realistically instead of being forced into one bucket.
- Richer marketing insight: membership scores can drive weighted or personalized campaigns (e.g., a user 60% "engaged" and 40% "spender" can receive a blended offer).
- Robust to noisy/borderline data: small behavioral changes cause gradual membership shifts rather than abrupt reclassification, which is more stable over time.
- Better reflects real-world behavior: user engagement is naturally a spectrum, not a strict category, so probability-based membership models this more faithfully.

For a mobile app company, soft clustering therefore provides a more nuanced, actionable view of user segments, supporting fine-grained retention and monetization strategies rather than a rigid one-size-fits-all classification.

Q4. Agricultural Crop Monitoring — Effect of Normalization on Clustering

Given Data

Farm	Rainfall (mm)	Soil Fertility Score	Crop Yield (tons)
F1	300	8	85
F2	500	4	70
F3	400	6	80
F4	280	9	90
F5	520	3	65

Clustering Before Normalization

Rainfall ranges from 280-520 mm while Soil Fertility Score ranges only 3-9 and Crop Yield ranges 65-90. Because clustering algorithms (e.g., K-Means) use Euclidean distance, the feature with the largest numeric range — Rainfall — dominates the distance calculation almost completely. As a result, farms get grouped mainly by rainfall (F1/F4/F3 as "lower rainfall" vs. F2/F5 as "higher rainfall"), while the genuinely important agronomic differences captured by soil fertility and yield are effectively ignored, even though fertility and yield vary meaningfully between farms.

Clustering After Normalization

After scaling all three features to a common range (e.g., Min-Max scaling to 0-1 or Z-score standardization), each feature contributes proportionally to the distance metric. The clusters that emerge now reflect the true agronomic pattern: F1, F3, and F4 (moderate rainfall, high fertility, high yield) form one cluster of "productive farms," while F2 and F5 (high rainfall, low fertility, low yield) form a second cluster of "less productive farms despite higher rainfall." This grouping is far more meaningful agronomically, since it shows that fertility — not rainfall alone — is strongly associated with yield.

How Normalization Improves Reliability

- Prevents any single large-scale feature (like rainfall in mm) from unfairly dominating the distance calculation.
- Ensures every variable contributes based on its actual pattern/variation, not its unit or magnitude.
- Produces clusters that align with domain knowledge (fertility and yield relationship), making insights more trustworthy and actionable for agricultural planning, such as recommending fertility-improvement interventions for low-yield farms rather than assuming rainfall is the only limiting factor.

Q5. Employee Performance Evaluation — Silhouette Score Analysis (8 Marks)

Given Data

Employee	Productivity Score	Attendance (%)	Skill Assessment Score
E1	88	92	84
E2	62	75	68
E3	78	88	72
E4	55	65	58
E5	93	96	90

Model (K)	Silhouette Score
K=2	0.48
K=3	0.68
K=4	0.52

Analysis of Silhouette Scores

The silhouette score measures how well-separated and internally cohesive clusters are, ranging from -1 (poor clustering) to +1 (dense, well-separated clusters). At K=2 the score is 0.48, indicating moderate structure but likely merging employees of noticeably different performance levels into overly broad groups. At K=3 the score rises to 0.68, the highest among the three models, indicating that three clusters produce the tightest, best-separated groupings for this data. At K=4 the score drops to 0.52, suggesting that splitting into four groups starts to fragment naturally cohesive clusters, reducing separation and cohesion — a sign of over-clustering.

Most Suitable Number of Clusters

K = 3 is the most suitable choice, since it has the highest silhouette score (0.68), the closest to +1 among the tested models, confirming clusters that are both compact and clearly separated from each other.

Justification and Business Implication

- Cluster 1 — High performers: E1 and E5 (high productivity, attendance, and skill scores) — candidates for leadership tracks, bonuses, and mentoring roles.
- Cluster 2 — Moderate performers: E2 and E3 (mid-range scores across all three metrics) — candidates for targeted upskilling and performance coaching.
- Cluster 3 — Low performers: E4 (lower scores across metrics) — candidates for structured improvement plans or closer supervision.

Because K=3 yields the highest silhouette score, it provides the most reliable and interpretable segmentation of employees. This directly supports effective performance evaluation by enabling HR to design differentiated interventions — recognition for high performers, development plans for moderate performers, and support plans for low performers — rather than relying on an undifferentiated (K=2, too coarse) or overly fragmented (K=4, less reliable) grouping.